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by B. Disson, R. Dussart, S. Iseni, et al.

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Humidity as a critical parameter for predicting breakdown voltage in sub-micrometer electrode gaps in air

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This experimental work re-investigates the breakdown voltage in air for electrode gaps ranging from 0.10 µm to 6.00 µm. Rarely addressed in the literature of the field, a special focus is placed on varying the gas humidity and its direct impact on the breakdown voltage. For short distances (< 2 µm), the results confirm significant deviations from the predictions of Paschen's law, with a linear increase of the breakdown voltage. The statistical approach of Disson *et al.* [1] reveals breakdown characteristics in dry air ($T_{dew} < -40\,°C$) to be very analogous to the previous results obtained in argon gas. There are –at least– two competing elementary mechanisms responsible to trigger the breakdown: the electrons produced by field emission (FE-mechanism) and the Townsend avalanche (A-mechanism). Breakdowns following the FE-mechanism occur systematically around the electric field value of $E_{FE} = 0.86\,V/nm$, which is very close to the value measured in argon. Unexpectedly, the gradual increase of humidity in air causes a counter-intuitive evolution of the breakdown voltages. While in macroscopic discharges, water molecules in the gas are known to elevate the breakdown voltages compared to dry gas, the admixture of H_2O drastically reduces the breakdown voltages in micro-gaps, e.g. 0.86 V/nm to 0.26 V/nm for dew points of $T_{dew} < -40\,°C$ and 10 °C respectively. The water layers adsorbed by the surface of the electrodes as well as the formation of a Taylor cone under the effect of the high electric field are discussed. This supports the analysis to understand whether H_2O enhances the FE-mechanism and A-mechanism or if this is an additional contribution leading to the breakdown. More generally, this study shed the light on the crucial role of humidity in micro-gap breakdowns and calls into question measurements results reported in the literature where humidity conditions are not documented.

I. INTRODUCTION

Humidity can greatly affect the mechanisms of discharge formation [2, 3], but surprisingly, unlike corona discharges or streamers [4], its impact on deviations from Paschen's law in microgaps has not been significantly considered and remains unclear. In other words, the role of water molecules in the breakdown for sub-micrometer distances has been hardly discussed. The purpose of this paper is to investigate breakdown voltages (V_b) in synthetic air (20.9 % O_2 ; 79.1 % N_2) with regards to considering the amount of water molecules, in order to bring the experimental conditions closer to those encounter in some electrical devices. This paper extends the newly introduced statistical approach [1] based on repeatability initiated in a previous article. In the latter paper, the histogram representation of V_b distributions in monoatomic rare gas (argon) revealed a competition between two breakdown mechanisms: the Townsend avalanche mechanism (A-mechanism) and the field emission mechanism (FE-mechanism). The method used for argon is applied to the case of dry air to investigate the case of molecular gas and to ensure that similar behavior occurs: an increase in the gap leading to a gradual transition of the dominant breakdown mechanism from FE-mechanism to A-mechanism. Due to the difficulty in estimating the impact of moisture on V_b, this paper proposes to properly reference humidity measurement in order to study its importance on the variations in breakdown mechanisms. Humidity seems much more complex due to its potential impact on electrode surfaces in addition to the gas phase. Furthermore, omitting the water vapor in the gas, also as residual impurities, must mislead the interpretation of the experimental data. This also a point to address at the available literature, where humidity was often not documented.

In this work, breakdowns in dry air are studied, before gradually increasing the water content diluted in the gas. Analyzing the evolution of breakdown voltage distributions in this context allows us to formulate hypotheses about the impact of water presence in the gas on breakdown mechanisms.

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II. EXPERIMENTAL METHOD

The experimental setup uses the work of Peschot *et al.* [5] and follows the recent study by Disson *et al.* [1]. Breakdowns are studied between a golden coated needle and a golden coated plane (Fig. 1). The previous study demonstrated that a tip radius of curvature of 20 µm is sufficiently large compared to the studied interelectrode distance range (0.10 to 6.00 µm) to approximate an homogeneous uniform electric field within the microgap in dry conditions [1]. An analytical calculation and a numerical simulation of the electric field show a field that is almost identical to a plane-plane configuration over a radial distance of 8.00 µm from the center of the needle. Each breakdown voltage measurement starts by the positioning of the electrodes using a piezoelectric actuator followed by the voltage increase at a rate of 10 V/s. Figure 1(a) shows a ballast resistor of 1 kΩ which limits

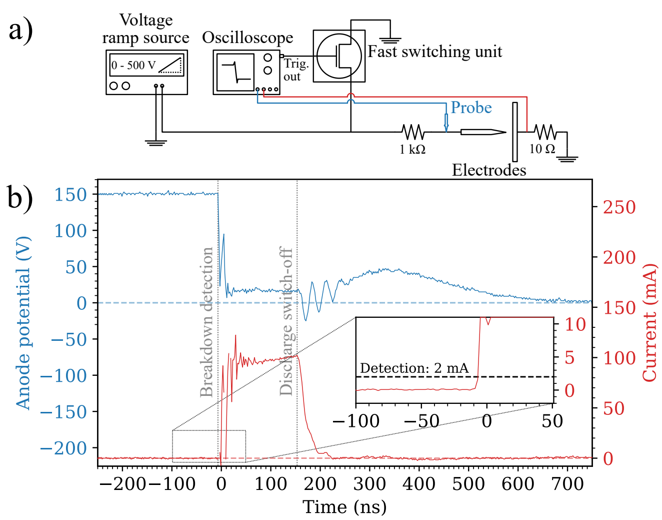

Figure 1. (a) Schematic diagram of the experimental discharge detection setup, and (b) evolution of voltage and current during a discharge in dry air.

the current in order to preserve the electrodes condition. As shown in Fig. 1, the current and voltage evolution are tracked during discharge on a nanosecond time scale using the oscilloscope. The use of a probe is necessary in order to limit the parasitic capacitance which could impact the breakdown voltage value. The criterion to consider a breakdown is based on the current and voltage waveform. It is essential noticing that Figure 1(b) highlights a voltage drop when the breakdown occurs. The sudden surge in current triggers the oscilloscope, sending a TTL signal to the cut-off system to reset the anode potential to 0 V. This happens in less than 200 ns as shown in Fig. 1(b). This setup enhances the repeatability, making the statistical analysis easier by increasing the number of measurements. Therefore, it is possible to perform a hundred breakdown voltage measurements for each experimental condition.

The amount of water vapor admixed in the air (humidity) is quantified by measuring the dew point (T_{dew}), *i.e.* the highest temperature at which the amount of water contained in the air would condense at fixed pressure [6]. Experimentally, T_{dew} value is measured by a Shaw probe. This information, combined with the gas temperature (T), can be used to calculate the relative humidity RH (ratio between the partial water vapor pressure (P_{H_2O}) and the saturated vapor pressure (P_{sat}) using Tetens' formula [7]. It is essential to note that the time interval between each measurement ($\sim$ 3 min) allows the thermal equilibrium between the gas and the electrode surface to be reached. Therefore, the gas temperature T is carefully recorded for each breakdown voltage measurement and most of the time the value is close to 25 °C. Although RH is commonly used to describe humidity in meteorology, this indicator can have the same value for two different gas compositions if the ambient temperature T is different. It is therefore preferable to use the molar fraction x_{H_2O} of water diluted in air (*i.e.* the ratio of water molecules in the mixing gas) which is an invariant quantity that does not depend on T. Rationally, this parameter makes more sense when studying physical mechanisms at the molecular level because it quantifies the proportion of water particles mixed in the gas where the discharge develops. It is possible to estimate x_{H_2O} from Magnus' formula which allows P_{sat} to be calculated independently of the total pressure P_{tot} [7, 8]. Consequently, multiplying RH by P_{sat} gives an estimation of the partial pressure of water vapor P_{H_2O} and the ideal gas law shows that the ratio between P_{H_2O} and P_{tot} is equal to x_{H_2O}. Consequently, x_{H_2O} can be expressed as a function of T_{dew},

$$x_{H_2O} = \frac{c}{P_{tot}} \cdot \exp\left(\frac{a}{b+T} \cdot \left(T + \frac{b \cdot (T_{dew} - T)}{b + T_{dew}}\right)\right). \quad (1)$$

Here, $a = 17.27$, $b = 237.7$ °C and $c = 6.1094 \times 10^{-3}$ bar are three constants [7]. x_{H_2O} value corresponding to the T_{dew} at 1.0 bar and with consideration of the ambient temperature T are reported in Table I. Controlling

Table I. Humidity level at which breakdown voltage measurements were performed. The experiment is considered done in dry condition if $T_{dew} < -40$ °C. The relative humidity RH is calculated from T_{dew} at experimental temperature and 1 bar using Tetens formula's [7]. Molar fraction of water mixed in the gas x_{H_2O} phase calculated with Eq. 1

T_{dew} (°C)	RH (%)	x_{H_2O} (%)
<−40 (dry)	<0.59	$<1.9 \times 10^{-4}$
−20	3.9	0.13
0	19	0.61
10	39	1.23

the humidity level during experiments is challenging because it requires replacing the impure atmospheric air with synthetic air (purity greater than 99.9999 %) and then adding water. To that end, a closed container containing sodium chloride moistened with distilled water is placed in the vacuum chamber. The walls of the recipient can let water pass through but not the solid salt. The ambient air trapped after closing the chamber is pumped down to 1×10^{-4} bar. Then the chamber is filled with dry synthetic air, until reaching the working pressure of 1.0 bar. The humidity has dropped drastically, reaching x_{H_2O} values lower than 0.05 % ($T_{dew} \leq -30\,°C$) but the gradual desorption of water contained in salt increases the amount of water vapor mixed in the air with time [9]. It is possible to control the humidity level at the desired T_{dew} value for the study by pumping and adding dry air periodically. Each series of measurements is preceded by a stabilisation phase to ensure that equilibrium between NaCl desorption, humidity absorption in the air and adsorption by metallic surfaces is properly reached. With this convenient method x_{H_2O} values greater than 1 % ($T_{dew} \geq 10\,°C$ at 25 °C) can be reached.

III. RESULTS AND DISCUSSION

Breakdown voltage measurements in air for different x_{H_2O} value are performed for distances ranging from 0.10 µm to 6.00 µm. The average values of hundred breakdown voltages for each experimental condition are shown in Fig. 2. As already observed in the literature [4], average V_b values in air deviate from the prediction of the Paschen's law through a quasi-linear section when the interelectrode gap becomes shorter and shorter. This phenomenon had already been observed in argon in the previous study [1] with an identical experimental setup. The main hypothesis to explain the deviations is related to the field emission quantum phenomenon. This mechanism, not considered by the classical Paschen's law, causes the electron emission from the cathode under a high electric field [11], leading in a drop of breakdown voltage [12, 13]. Figure 2 clearly shows this deviation regardless of the humidity level during the experiment. However, the curves clearly highlight the impact of humidity on average V_b values: the average V_b values decrease as moisture increases. Increasing humidity causes the plateau to begin at greater gaps, 0.50 µm in dry condition and 2.00 µm for $x_{H_2O} = 1.23\,\%$. This goes with a variation in the slope of the quasi-linear part: the slope is less steep if x_{H_2O} is higher. It is also important to note that the voltages on the plateau tend to decrease as humidity increases. For instance, a dew point of 10 °C (relative humidity of 38 %RH at 25 °C as shown in Table II) is a low value compared to what can be measured in the atmosphere at the Earth's surface [14]. Therefore, the appearance of the average breakdown voltage curve is similar to the ones obtained in early experiments conducted in atmospheric air [15, 16], as highlighted by the data from Germer [10] in Fig. 2. Although Germer used two palladium spheres as electrodes, his results are consistent with those of this paper because the work function of palladium metal (5.22 to 5.6 eV) is quite similar to that of gold (5.31 to 5.47 eV) [17]. Even the measurements taken by Torres and Dhariwal in atmospheric air conducted in a different electrode configuration show similar pattern [18, 19], considering that their experiments could be conducted in a more humid atmosphere. In general, measurements taken in atmospheric air lead to average breakdown curve shape that can vary considerably, as noted in [20]. These discrepancies could be explained by the varying humidity levels during the measurements. It is also important to note that the desorption of water from a metal surface is a timely process [21]. Pumping the chamber for several hours is often required to reach $T_{dew}\ le - 40\,°C$ (0.59 %RH) consider as dry conditions in this work.

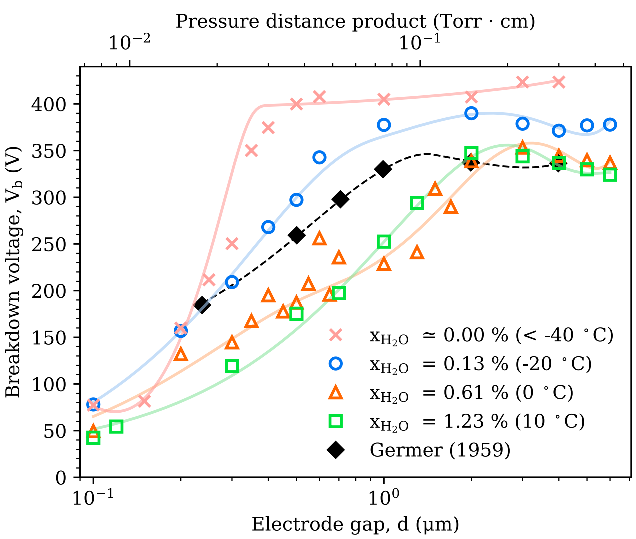

Figure 2. Average values of V_b in air at 1.0 bar for different x_{H_2O} value as a function of the interelectrode distance and the pressure distance product. The humidity is quantified by the molecular ratio and by the T_{dew} values, ranging from $< -40\,°C$ to 10 °C. The values obtained by Germer [10] are presented for comparison purposes and the trend lines is used as guide eyes.

However, Disson *et al.* [1] questioned the use of average V_b to study breakdowns in microgaps. The study of the total distribution via histograms is recommended when there is competition between several breakdown mechanisms. Figure 3 presents the distribution of a V_b series of 100 breakdowns in dry air for different gaps via three histograms. As observed in argon [1], the distributions also appear to be multimodal in dry air as it is possible to identify a narrow peak close to the Paschen's law prediction, and a broad peak explaining deviation observed in sub-micrometer electrode gaps. These two peaks also seem to evolve in a similar way to argon when the gap changes, *i.e.* the narrow peak taking over when the interelectrode distance increases. Fitting histograms of V_b distribution with the method explained in [1] provide information about the positions of the peaks and makes possible to quantify the proportion of breakdown following A-mechanism and the proportion of breakdown

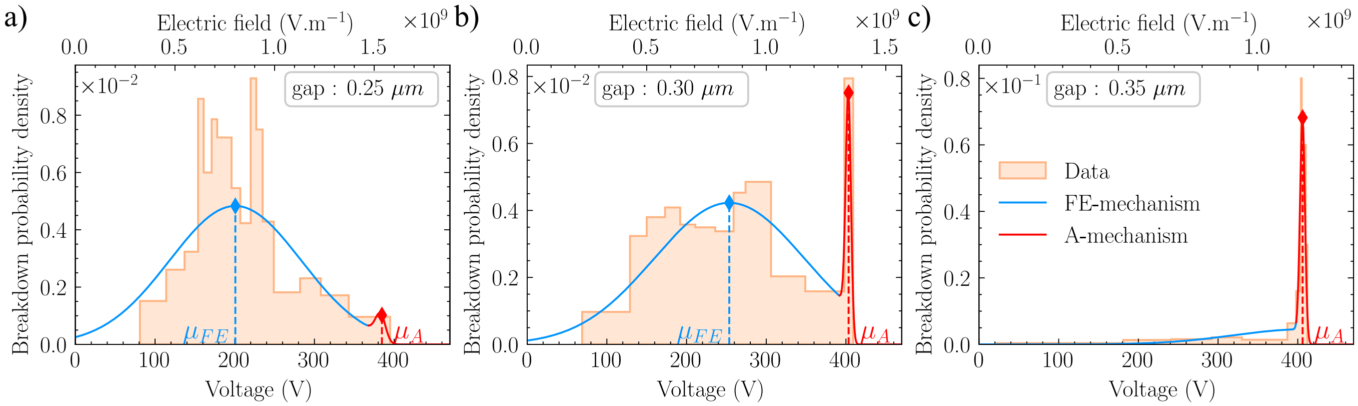

Figure 3. Measured distributions of a series of 100 breakdowns for three different distances in dry air ($T_{dew} < -40\,°C$): a) 0.25 µm, b) 0.30 µm and c) 0.35 µm. The probability density function from [1] is fitted on the histograms.

following FE-mechanism. It describes the narrow mode and the broad mode at respective positions μ_A and μ_{FE} and with respective dispersion coefficients σ_A and σ_{FE}. However, it is worth noticing that the transition from a narrow peak to a broad peak occurs for larger gaps in dry air (arround 0.30 µm) than in argon(arround 0.20 µm). Also, the μ_A values (around 400 V) are also greater than the one in argon (around 250 V), which is consistent with the overall trend of the Paschen's law [22, 23], and so with the ionization cross section [24]. It should be noted that the broad peak was always positioned around the same electric field values (μ_{FE} close to 8.9×10^8 V/m) [1]. Interestingly, this appears to also apply in dry air, reinforcing the idea that the breakdowns occurring around μ_{FE} values are related to field emission. It suggests that the FE-mechanism is unlikely to be gas-dependent [25].

Figure 4 shows the evolution of the probability density functions [1] obtained from the histograms fitting in dry air at 1.0 bar versus the interelectrode distance. This representation clearly highlights the gradual transition between the two breakdown processes, with the appearance of the singular peak attributed to the Townsend mechanism above a certain gap threshold (250 nm). The latter develops in proportion as the gap increases. For larger gaps (> 1 µm), the broadening of the mode is explained by the weakening of the in field emission while the pure avalanche –with its inherent collisionnal properties– is driving the discharge.

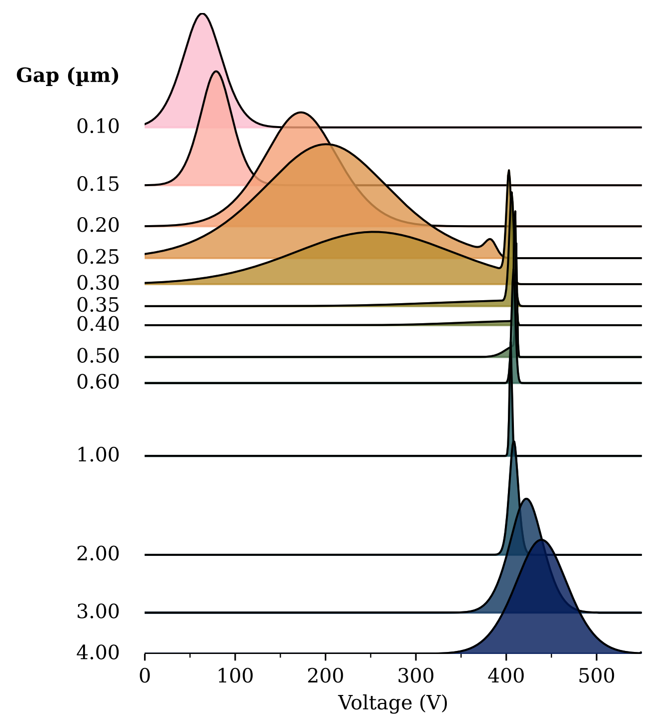

Figure 4. Evolution of the breakdown probability density versus the interelectrode distance in dry air ($T_{dew} < -40\,°C$) at 1.0 bar.

Figure 5(a) shows the evolution of μ_A, μ_{FE}, and the complete distribution of V_b values, approximated using kernel estimation [26]. It is important to note that the value of E_{FE} for dry air is 0.86 V/nm, which is close to 0.77 V/nm, the value found for argon by Disson *et al.* [1]. This observation supports the idea that, in the absence of water, the FE mechanism is independent of gas. The same measurement procedure is repeated for different x_{H_2O} values ranging from $\sim 0\,\%$ (<1.9 ppm) to 1.23 % and the result are synthesized on violin diagrams in Fig. 5. In humid air, the distribution of breakdown voltages remains bimodal. It is therefore highly probable that adding water to the gas phase will not cause a third breakdown mechanism to occur. However, Figure 5 clearly shows differences in distribution, whether for the mode related to field emission mechanism or for the mode related to the avalanche mechanism. Table II summarizes the experimental conditions T_{dew}, H_2O and the electric field values E_{FE} corresponding to the slope coefficient shown in Fig. 5. This data highlights a decrease of the slope coefficient (in V/nm) as the humidity increases, but also a gradual loss of linearity resulting in a reduction of μ_{FE} values as humidity increases. Therefore,

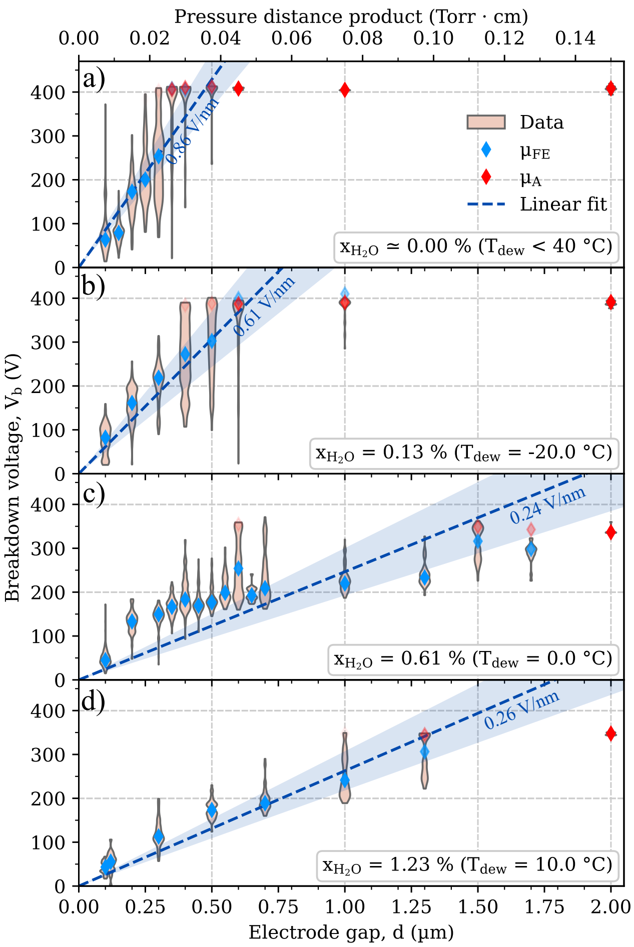

Figure 5. Violin plot representing V_b measurements made in dry air (a) and humid air with $\mathrm{T_{dew}}$ values of $-20\,^\circ$C (b), $0\,^\circ$C (c) and $10\,^\circ$C (d) for distances ranging from 0.10 µm to 2.00 µm. The fitted parameter μ_A and μ_{FE} are represented by diamonds with a gradual transparency defined by the weighted values of w_A and w_{FE} [1]. A linear fit of μ_{FE} and its 2σ-confident interval is presented.

the proportional model performed less and less well μ_{FE} as the moisture increases, explaining the modest raise in E_{FE} (and so h_{max}) for $\mathrm{T_{dew}} = 10\,^\circ$C despite an overall decreasing trend. A lower E_{FE} means that breakdowns

Table II. Electric field values obtained by linear regression in Fig. 5 and approximation of the Taylor cone maximum height possible under the stability condition, calculated using Eq. 2.

$\mathrm{T_{dew}}$ (°C)	E_{FE} (V/nm)	h_{max} (nm)
<−40 (dry)	0.86	19.0
−20	0.61	37.1
0	0.24	230.4
10	0.26	202.9

following the FE-mechanism occur around an electric field value that decreases as humidity increases. Considering this mechanism to be independent of the gas phase, it appears the water vapor admixed in the gas phase should not directly affects the μ_{FE} values. Then change in field value most likely reflects a change in the surface properties of the electrodes because of the adsorbed water presence in liquid form on the surface of the electrodes. Meanwhile, the increase in water content within the gas phase is inherently accompanied by water adsorption on the surface [21, 27, 28]. The thickness of the water layer on the walls depends not only on the humidity level, but also on many parameters, such as the material, roughness, and contact angle [21, 27]. It is worth noticing that while the x_{H_2O} parameter can be used to describe the gas composition independently of the ambient temperature, the Brunauer–Emmett–Teller theory describes the adsorbed layer thickness on the surface of a material as being highly dependent on it [29, 30]. However, measurements of water mass gain give an estimation of the water layer thickness , which can be as thick as a few nm on gold, depending on the relative humidity [21, 27, 28, 31]. It is noteworthy to mention, the presence of water in contact with the metal leads to a decrease in work function [27, 32], which plays a critical role by increasing field emission from the cathode, thereby enhancing breakdown through the FE-mechanism.

Moreover, the water layer can form a protrusion shape-like under the effect of high electric fields, to the point of forming a Taylor cone if the electric field is sufficiently high as illustrated in Fig.6. Although this phenomenon is mainly documented and investigated on the macroscopic scale [33, 34], deformations have already been studied for water thicknesses of 20 nm [35]. It was first described empirically by Taylor [36] by assuming that its formation is a consequence of a balance between electrostatic pressure and inward pressure caused by the surface tension [37]. This gives a practical and very simple estimation of the Taylor cone shape,

$$\frac{1}{2} \cdot \epsilon_0 \cdot E_c = \frac{\gamma}{h_{max} \cdot \tan(\theta_T)}. \qquad (2)$$

In Eq. 2, γ is the surface tension of the water (71.99×10^{-3} N/m [38]), ϵ_0 is the vacuum dielectric permittivity (8.85×10^{-12} F/m), h_{max} is the maximum height of the cone and E_c is the critical electric field above which the cone loses its stability. θ_T is the semi-vertical angle of the cone, with a constant value of 49.3° as demonstrated in the work of Taylor [36]. Although this value may vary depending on space charge, flow rate, and ambient pressure [34, 39], this angle value is close to what can be found experimentally and with numerical simulations [40, 41]. Equation 2 gives an estimation of the maximum size of the Taylor cone, imposed by the stability condition. The values given in Table II correspond to an upper limit of the actual height (h) of the deformation ($h \leq h_{max}$, as shown in Fig. 6), considering E_{FE} as the most probable critical electric field values. h_{max} calculated by the stability condition may be greater than the interelectrode distance because of the lack of precision of this empirical approach which is only a simplified version of reality [34, 42]. However, the order of magnitude highlights that the gap reduction due to the deformation of the water layer may have a significant impact on the effective gap size ($d_{H_2O} \leq d$), especially if the humidity level is high. The size and shape of Taylor's cone depends on the amount of water available on the electrodes, the charge accumulation and the significant local enhancement of the electric field (E_{H_2O}) due to the deformation. A numerical approach is required for a precise estimation [35, 43]. It is important to note that a stronger electric field leads to an increase in h [33] but not in h_{max} [44]. This can be explained because the actual shape of the water deformation increasingly approaches the maximum conical shape as the electric field increases. As observed with a similar electrode geometry [45], the height of the stable conical shape h_{max} decreases as E_0 increases.

All these local changes on the surface and in the electrical properties lead to a deterioration in homogeneity and uniformity of the electric field that could explain the gradual loss of linearity in the evolution of μ_{FE} when x_{H_2O} increases.

It is worth noticing that at the sub-micrometer scale and under a high eletric field, the pressure exercised by the gravity ($P_g = \frac{1}{3} \cdot h_{max} \cdot \rho \cdot g \approx 3 \times 10^{-4}$ Pa) is negligible compared to the electrostatic pressure ($P_e = \frac{1}{2} \cdot \epsilon_0 \cdot E_c^2 \approx 3 \times 10^6$ Pa) and the pressure caused by the surface tension ($P_\gamma = \frac{\gamma}{h_{max} \cdot \tan(\theta_T)} \approx 3 \times 10^6$ Pa) [37]. Consequently, gravity, which prevents the formation of two liquid-covered surfaces facing each other, has very minimal impact on the shape of sub-micrometer scale water layer. The hypothesis of the presence of two Taylor cones facing each other, as outlined in Fig. 6 is then to be considered. In addition, Bruggeman *et al.* [33] mentioned that applying a positive or negative potential to the tip (*i.e.* using it as an anode or cathode) does not cause any major differences on the breakdown voltage values, and so on the water layer deformation. Consequently, it is reasonable to consider the hypothesis of the water layer deformation on the anode and on the cathode, as shown in Fig. 6.

Finally, various studies on macro-gaps have shown a moderate increase in V_b with humidity [46, 47]. This can be explained by the greater attachment of electrons in humid air than in dry air [46], or by the ability of water molecules present in humid air to recombine very quickly after dissociation [47]. Increased humidity would result in difficulties to trigger and maintain discharge via Townsend avalanche (A-mechanism), which would cause V_b to increase. However, the study conducted in this paper tends to show the opposite trend, with lower μ_A values when x_{H_2O} is high. This is similar to the results obtained by Bruggeman *et al.* [33], who observed in larger gaps a drop in breakdown voltage when using a water electrode. Above a critical electric field value, the Taylor cone formed by the moving surface begins to emit ions and subsequently water particles [36, 42, 48]. The emission process, also known as electrospray, is already well documented at the macrocroscopic scale [49–51] and at microscopic scale [42, 48]. Moreover, the particle emission process has already been studied in the case of field ion microscopy where it had already been noted that water adsorbed on the surface of the emitter tips strongly affects particle emission. [52–54]. It has been studied by Rothfuss *et al.* [54] who show the formation of a protrusion on the metal surface which locally decreases the radius of curvature and increases the electric field. Most of the time, the anode is used as the emission electrode, but the particle emission can also occur from the cathode. Bruggeman *et al.* [33] observed that V_b measurements follows critical voltage curve, the particle emission triggering breakdown at lower voltage values [33]. This phenomenon can be observed if the pre-breakdown electric field is capable of reaching the critical value, which could possibly happen in microgaps due to the short distance between electrodes generates high electric field. This allows ions to be emitted from the water layers, causing breakdowns at lower μ_A values in humid air. However, it is important to note that the particles emitted depend on many parameters but specifically the ambient temperature, which could potentially explain the observable discrepancies from one experiment to another that may be conducted at similar humidity levels but at different temperatures [53].

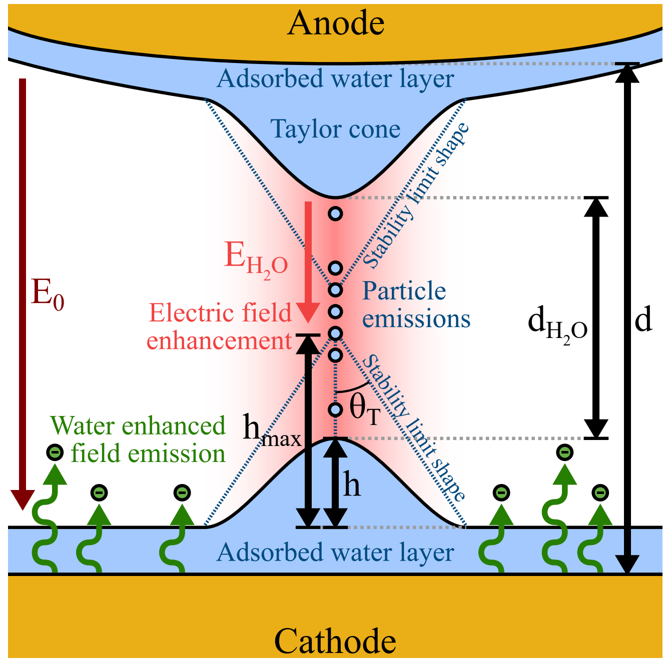

Figure 6. Schematic diagram –not to scale– presenting the impact of a electric field E_0 on the adsorbed water layer and on the formation of sub-micrometric discharges, through field emission enhancement and electrode geometry modification via Taylor cone formation. The deformation of the adsorbed water layer causes a local enhancement of the electric field ($E_0 \leq E_{H_2O}$) but also a reduction in the effective interelectrode distance($d_{H_2O} \leq d$). The changes in the discharge condition lead to modification in breakdown mechanisms and consequently to a drop in V_b.

IV. CONCLUSION

In conclusion, this study presents results that corroborate the observations and hypotheses arising from the new method for studying breakdown voltages proposed in [1] and based on statistics. Firstly, the experiments conducted in dry air revealed the same tendency as the results previous results obtained with argon, (*i.e.* a linear increase of V_b with the inter-electrode distance for rather short gaps and a bi-modal distribution. It is therefore reasonable to assume that, as with argon, breakdowns in dry air for small interelectrode gaps are subject to competition between two breakdown processes: the field emission mechanism and the Townsend mechanism (avalanche). Furthermore, breakdowns following the A-mechanism reach values close to those predicted by Paschen's law, with V_b values higher in air than in argon. Breakdowns following the FE-mechanism systematically appear around the same electric field value: 0.86 V/nm. This value is very close to the one obtained in argon (*i.e.* 0.77 V/nm [1]), which supports the idea that this mechanism does not depend on the gas phase.

It could be intuitive to think that V_b values would increase with humidity for nanometric discharges similar to discharges in macrogaps. Unexpectedly, the measurements clearly revealed the opposite: in very small gaps, increasing the humidity leads to significantly reduce the breakdown voltage. These results report the importance of studying the complete distribution of breakdown voltages in order to better understand how humidity affects breakdown mechanisms. As already mentioned, two distinct mechanisms leading to breakdown have been identified. However, the role of water molecules is rather complex because it is not clear whether the humidity is affecting one mechanism or the other –or both. Yet, one does not exclude another intrinsic contribution of water molecule to support the field emission. Indeed, the evolution of breakdown voltage distributions reveals that the presence of water affect the effective V_b values while operating the experiment in the field emission regime. Breakdowns following this mechanism occur around an electric field values that can substantially drop from 0.86 V/nm in dry air ($T_{dew} < -40\,°C$) to 0.24 V/nm in humid air ($T_{dew} = 10\,°C$). On a sub-micrometer scale, the presence of moisture on the electrode surfaces cannot be ignored. From the literature, it is suggested that the layer of adsorbed water reduces the work function, which increases the field emission and would explain the drop of V_b for breakdown following FE-mechanism.

Additionally, it is possible that the high electric field of microgaps causes a deformation of the water surface. A known liquid surface modification under intense electric field is the formation of the Taylor cone. This would change the initial planar surface geometry of the electrodes, reducing the interelectrode distances and significantly enhance the electric field locally. Above a critical electric field threshold, Taylor cones act as nanoelectrosprays, potentially emitting ions and water particles, capable of triggering avalanche breakdowns at lower voltages.

Last but not least, this study calls into question literature data on micro- nano-gap breakdown, where humidity conditions are not documented. Investigating the impact of humidity on breakdowns and the subsequent reduction in V_b could unlock new applications, such as the development of ultra-low-voltage electric spark gaps.

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